Problem-Driven Maritime Logistics for Ghana: Cryptographic Consignment Auditing and Port Clearance Optimization for Tema and Takoradi Ports

BERNARD FIAGBENU
Computer Scientist · [Affiliation]
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ABSTRACT

Maritime cargo clearance through Ghana's primary ports serves as the vital commercial conduit for both the domestic economy and landlocked Sahelian neighbors. This paper investigates the top five supply chain bottlenecks in Ghanaian port logistics and develops an immutable cryptographic consignment audit ledger extending China Xpress. We evaluated logistics performance indicators from the Ghana Ports and Harbours Authority (GPHA), Ghana Revenue Authority (GRA), and World Bank Logistics Performance Index (LPI), ranking candidate bottlenecks using formal multicriteria scoring. This paper targets the five critical supply chain bottlenecks in Ghana: container demurrage disputes at Tema Port, customs document forgery, maritime shipment tracking opacity, SME import trade finance barriers, and inter-agency digital silos. As a tangible open-source artifact, we engineered and verified a cryptographic consignment manifest audit ledger module in /prototypes/supply-chain/ extending the operational China Xpress platform.

Keywords: Maritime Logistics, Tema Port, Takoradi Port, China Xpress, Cryptographic Auditing, ICUMS, Demurrage Reduction

PROBLEM-TO-SOLUTION SUMMARY & OPEN ARTIFACT

This publication presents an empirical pathway connecting established computational and engineering theory to physical, field-validated artifacts addressing five ranked national problems in Ghana. A runnable, tested open-source prototype has been authored and published under an MIT license in the repository directory /prototypes/supply-chain/ with automated unit test suites.